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Understanding the Supramolecular Chemistry of Lutein with Gelatin, Octenyl Succinic Anhydride-Modified Starch Polymers in Simulated Gastro-Intestinal Buffers: A Molecular Dynamics Study

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Lutein, a carotenoid with significant health benefits, faces challenges in delivery due to its poor solubility and low bioavailability in the human body. Experimental methods are used to deliver lutein but it has limitations in providing detailed molecular-level insights into lutein interactions within delivery systems. Molecular dynamics (MD) simulations offer a powerful tool to overcome this knowledge gap by providing an atomic-level understanding of lutein behavior in various physiological conditions. This study employs MD simulations to investigate the binding of lutein in a hydrogel composed of gelatin and octenyl succinic anhydride (OSA)-modified starch across different gastro-intestinal buffers. The analysis of the simulation trajectory reveals that while both gelatin and OSA-starch form hydrogel networks, OSA-starch maintains better structural integrity in all buffers. Autodock scores indicate a higher binding affinity of lutein for OSA-starch compared to gelatin, which is further validated by hydrogen bonding and radial distribution function analyses. The mean square displacement results demonstrate the subdiffusive behavior of lutein. These results provide valuable insights into the interactions of lutein with hydrogel components and solvents, contributing to the rational design of improved lutein delivery systems. Future research can focus on optimizing the hydrogel composition to enhance lutein stability and bioavailability in various physiological environments.

1. Introduction

Lutein, a xanthophyll carotenoid, plays a crucial role in the medical industry due to its numerous health benefits. It is widely recognized for its potent antioxidant properties and its ability to protect the eyes from harmful light and oxidative stress.[1–3] Lutein accumulates in the macula of the human eye, where it acts as a natural filter against blue light and helps maintain optimal visual function. Research indicates that lutein supplementation may reduce the risk of age-related macular degeneration, cataracts, and other ocular disorders.[3–5] Beyond eye health, lutein demonstrates promising effects in cardiovascular health,[6] cognitive function,[7] and skin protection.[8] These diverse applications make lutein a valuable compound in both preventive and therapeutic medical contexts. Despite its potential health benefits, the consumption of free lutein presents several challenges. Lutein is highly lipophilic and exhibits poor water solubility, which significantly limits its bioavailability when consumed orally.[9,10] This low solubility leads to poor absorption in the gastrointestinal tract, reducing its effectiveness. Additionally, free lutein is susceptible to degradation when exposed to light, heat, and oxygen, further compromising its stability and efficacy.[10,11] These limitations necessitate the development of effective delivery systems to enhance lutein's bioavailability and protect it from environmental factors. To overcome these challenges, various methods have been explored to improve lutein delivery. Encapsulation techniques, such as nanoencapsulation and microencapsulation, have shown promise in enhancing lutein stability and bioavailability.[12,13] Emulsion-based systems, including nanoemulsions and microemulsions, offer another approach to increase lutein solubility and absorption.[9,14] Liposomal formulations have also been investigated as potential carriers for lutein.[15] Among these methods, hydrogel-based delivery systems have emerged as a particularly promising approach for lutein administration.[16] Hydrogels possess several advantageous properties that make them suitable carriers for lutein. Their 3D network structure allows for high water content while maintaining structural integrity, providing a favorable environment for hydrophobic compounds like lutein. Hydrogels can be designed to respond to various stimuli, such as pH, temperature, or enzymatic activity, enabling controlled release of the encapsulated lutein. Furthermore, hydrogels can be formulated to protect lutein from degradation during storage and gastrointestinal transit, enhancing its stability and bioavailability.[17,18]

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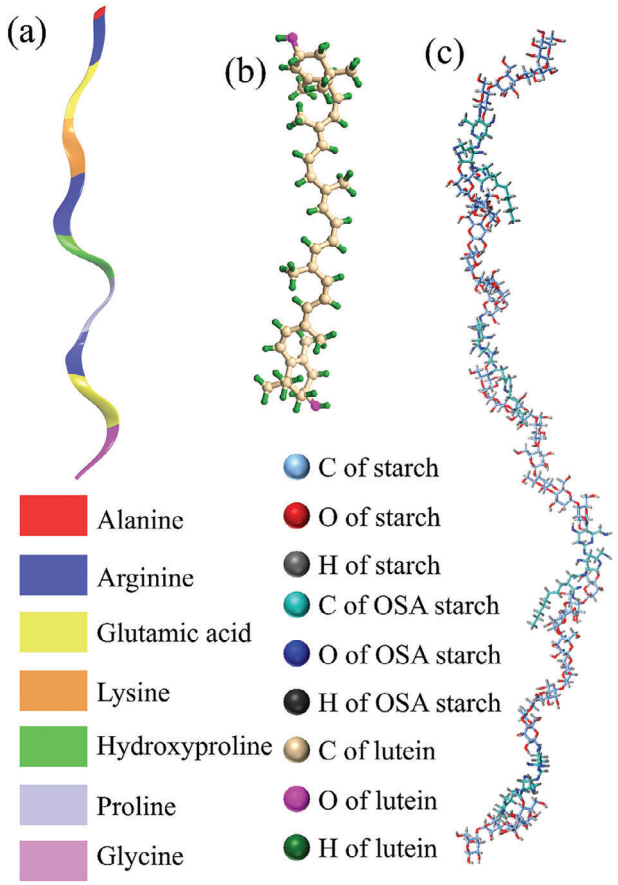


Figure 1. Initial structure of polymer framework and drug a) gelatin b) lutein c) OSA-modified starch.

There are several hydrogel formulations available that carry lutein.[16–19] Among them, the gelatin modified with octenyl succinic anhydride (OSA)-modified starch has garnered significant attention as a potential carrier for lutein.[19] Gelatin, a protein-based biopolymer derived from partial hydrolysis of collagen, is widely used in the food and pharmaceutical industries due to its biodegradability, biocompatibility, and ability to form thermoreversible gels.[20,21] Gelatin hydrogel has several advantages, but it also has several disadvantages, such as poor mechanical strength, thermal instability, and relatively short degradation time.[22,23] To overcome these challenges, the gelatin polymer blends with the OSA-modified starch, improving the emulsification properties and hydrogel system stability. This hydrogel system offers a synergistic approach to lutein delivery, leveraging the strengths of both components.[19,24] This hybrid system not only addresses the solubility and stability issues associated with free lutein but also offers the potential for controlled release and targeted delivery. The gelatin and OSA-modified starch-based hydrogel creates a protective environment for lutein, shielding it from degradation factors such as light, heat, and oxidation. Despite the promising potential of gelatin modified with OSA-starch hydrogels for lutein delivery, several knowledge gaps persist in experimental studies. These include a limited understanding of the precise molecular interactions between lutein and the hydrogel components, the lack of detailed information on the binding sites and mechanisms, and insufficient data on how different environmental conditions affect the hydrogel's performance. Hydrogen bonding is a crucial property of the characteristics of the hydrogel, which is not quantified using experimental data. To address these knowledge gaps, molecular dynamics (MD) simulations offer a powerful computational approach. MD simulations can provide atomic-level insights into the interactions between lutein and the hydrogel components, revealing the binding sites and mechanisms that are challenging to observe experimentally.[25–27]

MD simulations have emerged as a powerful tool in computational pharmaceutics, offering atomic-level insights into drug delivery systems and molecular interactions that are often challenging to observe experimentally.[26,27] These simulations have been successfully applied to study various aspects of pharmaceutical research, including the behavior of hydrogel-based drug delivery systems, protein-ligand interactions, and enzyme inhibition.[28–31] In hydrogel studies, MD simulations elucidate how factors such as ionic strength and pH affect water states, mesh size, and mechanical properties, providing a deeper understanding of drug release kinetics and hydrogel stability. For protein-ligand interactions, MD simulations offer valuable information on binding mechanisms, thermodynamic parameters, and conformational changes, as demonstrated in studies involving β-lactoglobulin and natural compounds like rutin, resveratrol, and curcumin.[26,32] In enzyme inhibition research, such as investigations into matrix metalloproteinase 9 (MMP-9), MD simulations explore binding modes of potential inhibitors, assessing their stability and identifying key amino acid residues involved in interactions.[28] The integration of MD simulations with other computational techniques, including molecular docking and multivariate statistical methods, has led to more comprehensive approaches to predicting drug loading and formulation properties, potentially reducing the time and cost associated with experimental trials. As computational power continues to increase, MD simulations are becoming an increasingly important tool in pharmaceutical research, bridging the gap between experimental observations and theoretical predictions, and contributing significantly to the advancement of computational pharmaceutics and the development of more effective drug formulations.

This study employs molecular dynamics (MD) simulations to investigate the binding interactions of lutein within a composite hydrogel system composed of gelatin and OSA-modified starch. The aim is to elucidate the specific binding sites and non-bonded interactions of lutein in three physiologically relevant environments: phosphate buffer solution (PBS), artificial gastric fluid (AGF), and artificial intestinal fluid (AIF). By utilizing MD simulations, the research seeks to provide atomic-level insights into the behavior of lutein within this hydrogel system across different physiological conditions. Various parameters including hydrogen bonding, radial distribution functions, and mean square displacements are analyzed to characterize the interactions between lutein and the hydrogel components.

2. Computational Methodology

The initial structure of gelatin, OSA-modified starch, and lutein are found in the literature.[33,34] The Avogardo open-source package is used to draw the gelatin, monomeric structure of

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Table 1. Details of the systems.

Buffer solution	PBS	AIF	AGF
Initial box dimensions [nm^3]	50 × 50 × 50	50 × 50 × 50	50 × 50 × 50
Final box dimensions [nm^3]	10.3 × 10.3 × 10.3	10.3 × 10.3 × 10.3	10.3 × 10.3 × 10.3
Final density [Kg m^{-3}]	1211.78	1207.54	1209.42
Number of Gelatin atoms	16 008	16 008	16 008
Number of OSA-modified Starch atoms	40 950	40 950	40 950
Number of Lutein atoms	196	196	196
Number of H_3O^+ atoms	–	–	128
Number of Na^+ atoms	65	–	12
Number of Cl^- atoms	116	116	160
Number of K^+ atoms	–	10	–
Number of $H_2PO_4^-$ atoms	63	70	–
Number of HPO_4^{2-} atoms	168	–	–
Number of Water atoms	80 000	80 000	80 000
Total Number of atoms	137 461	137 350	137 454

the starch, and lutein structure.[35] The monomeric structure of starch replicates five times, as explained in the literature.[25] The third monomer of oligomeric starch structure is functionalized with octenyl succinic anhydride to avoid steric hindrance and generate the OSA-modified starch. Further, the OSA-modified oligomeric starch replicates four times to generate the polymeric starch framework. The initial structures of macromolecules and lutein are given in **Figure 1**.

AutoDock is employed to investigate the binding interactions between a polymer and a drug after preparing the polymer structure. First, the polymer and drug structures are prepared by adding the polar hydrogen atoms and assigning partial charges. Next, grid box dimensions are defined to encompass the potential binding site on the polymer surface. A set of random drug conformations is generated within the grid box, and their binding energies are evaluated using the AutoDock scoring function. Drug conformations with the lowest binding energies are selected as the most likely binding poses.

After docking the drug moiety in the polymer framework, the MD simulations are used to explore the dynamics of lutein loaded with macromolecules in the phosphate buffer solution (PBS), artificial gastric fluids (AGF), and artificial intestine fluids (AIF). The compositions of PBS, AGF, and AIF are taken from the literature.[36] The appropriate amount of monosodium phosphate and disodium phosphate are mixed in water to create the PBS solution. A suitable amount of hydrochloric acid and sodium chloride are added to the water to create the AGF solution. The AIF solution is created to mix the water and dipotassium phosphate. The TIP4P water model is used to create the PBS, AGF, and AIF solutions.[37] The TIP4P water model includes four atoms per molecule: one oxygen, two hydrogens, and one virtual site representing the lone pair of oxygen atoms. TIP4P provides excellent agreement with experimental density at 25 °C and 1 atm, accurate structural data, and good thermodynamic properties.[37] After the creation of PBS, AGF, and AIF, the Packmol open-source package is used to immerse the lutein-loaded macromolecules and generate the initial configurations. Detailed information about the systems is given in **Table 1**. To understand the system's dynamics, the force field is required to compute the interactions. The bonded and non-bonded interactions are computed using the Amber14sb forcefield. The functional form of Amber14sb forcefield is given in the literature.[38] To perform the dynamics, the Gromacs-2022.3 open-source MD package is used.[39]

The system's dynamics are obtained through energy minimization, equilibration, and production run. The steepest decent algorithm is employed with a cut-off value of 1000 kJ $mol.^{-1}$ nm and the maximum number of steps of half a million for energy minimization. The cut-off distances for van der Waals (vdW) and electrostatic interactions are 1.0 and 1.2 nm, respectively. The long-range electrostatic interactions are computed using the Particle Mesh Ewald algorithm.[40] The periodic boundary conditions are applied in all three dimensions. The leapfrog integration algorithm is used to compute the Newton's equation of motion. The equilibration is performed in two stages NVT and NPT equilibrations. The NVT equilibration is performed for five ns with a time step of two fs. The system's temperature is maintained at 310 K with the help of a velocity-rescaling thermostat. The

Table 2. Auto dock score for gelatin and lutein.

Mode	Affinity [Kcal mol^{-1}]	Distance from Best Mode	
		rmsd l.b.	rmsd u.b.
1	−3.2	0	0
2	−3	10.437	14.38
3	−2.9	2.237	7.169
4	−2.9	9.921	12.839
5	−2.8	10.657	14.268
6	−2.8	4.28	8.899
7	−2.8	2.046	4.282
8	−2.8	9.193	13.185
9	−2.8	2.307	7.577

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Table 3. Auto dock score for OSA-modified starch and lutein.

Mode	Affinity [Kcal mol^{-1}]	Distance from Best Mode	
		rmsd l.b.	rmsd u.b.
1	−4.2	0	0
2	−3.9	36.577	40.343
3	−3.8	37.305	40.677
4	−3.8	37.982	41.416
5	−3.7	37.232	40.741
6	−3.6	36.504	40.19
7	−3.6	36.837	40.206
8	−3.6	35.612	39.176
9	−3.5	36.055	39.457

NPT ensemble performed for 45 ns with a time step of 2 fs. During the NPT equilibrations, the system's temperature and pressure are maintained at 310 K and 1.013 bar. The velocity-rescaling thermostat[41] and Parinello-Rahaman[42] barostat are used to maintain the system's temperature and pressure. During equilibration, the system's hydrogen-containing moiety motion is frozen using the LINCS algorithm.[43] The production run is performed for 50 ns with a time step of 2 fs. During the production run, the system's temperature and pressure are maintained at 310 K and 1.013 bar with the help of a Velocity-rescaling thermostat and Parinello-Rahaman barostat. The system's trajectory is saved every 10 ps interval for statistical averaging and data analysis. The data visualization is carried out in VMD,[44] UCSF Chimera-X.[45,46] The analysis is done using Python in-house coding.

2.1. Trajectory Analysis

2.2. Hydrogen Bonding (H-Bonding)

The H-bonding is an attractive interaction between a hydrogen atom bonded to a highly electronegative atom (such as oxygen or nitrogen) and another electronegative atom. MD simulations can quantify hydrogen bonds by analyzing the distances and angles between atoms that meet specific criteria. The H-bond is considered to exist when the distance between the donor and acceptor atoms is below 3.5 Å, and the angle between the donor-hydrogen-acceptor is less than 30°. The mathematical form of H-bonding is found in the literature.[47,48]

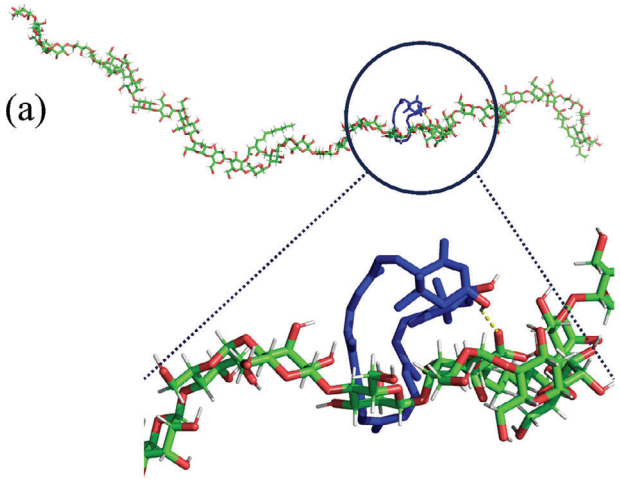


Figure 2. Molecular docking of lutein in OSA-modified starch.

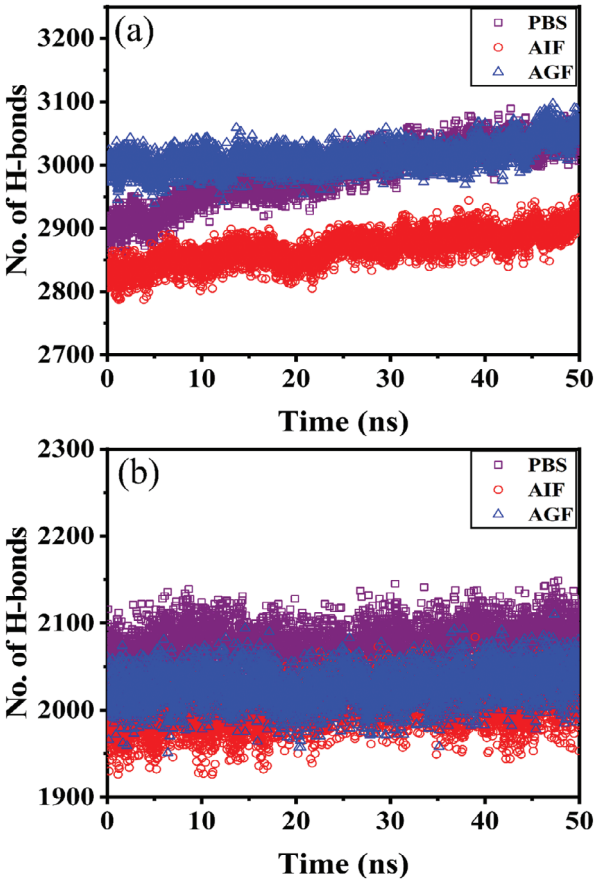


Figure 3. H-bonding between the polymer to the solvents a) gelatin to solvents b) OSA-modified starch to solvents.

2.3. Radial Distribution Function (RDF)

The RDF, also known as the pair correlation function, is a key statistical tool in MD simulations that describes how the density of particles varies as a function of distance from a reference particle. It provides information about the structure of the system at the atomic level. The mathematical form of RDF is given in Literature.[49]

2.4. Mean Square Displacements (MSD)

MSD is a measure of the average distance that particles in a system travel over time. In MD simulations, it is calculated by tracking the positions of particles at different time steps and computing the square of the displacement from their initial positions. The MSD is then averaged over all particles and multiple time

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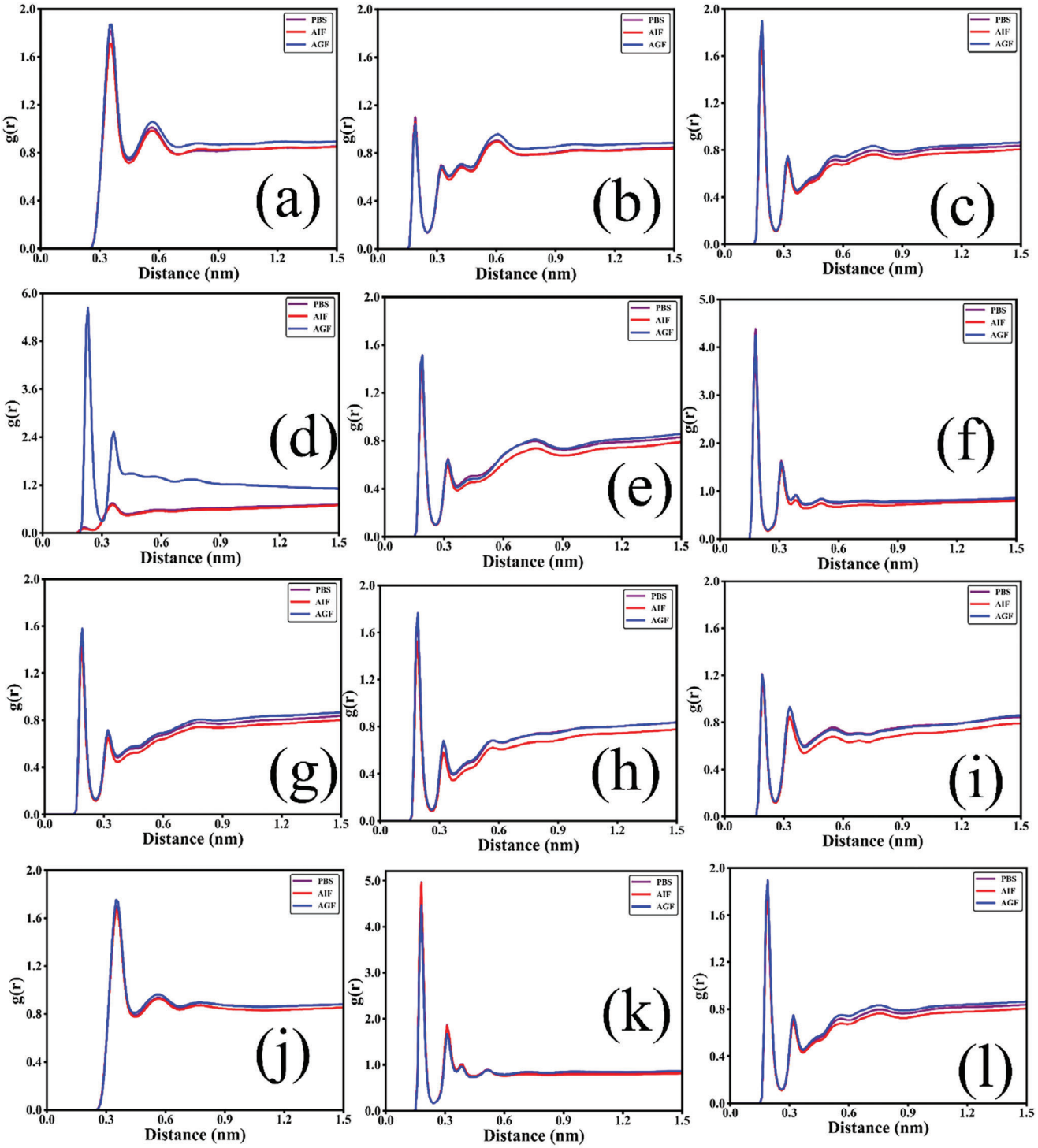


Figure 4. RDF between the different moiety of gelatin to the hydrogen of water a) N of alanine to H of water b) O of alanine to H of water c) O of arginine to H of water d) Cl to H of water e) O of glutamic acids to H of water f) OE1 & OE2 of glutamic acids to H of water g) O of glycine to H of water h) O of hydroxyproline to H of water i) OD of hydroxyproline to H of water j) N of lysine to H of water k) O of lysine to H of water l) O of proline to H of water.

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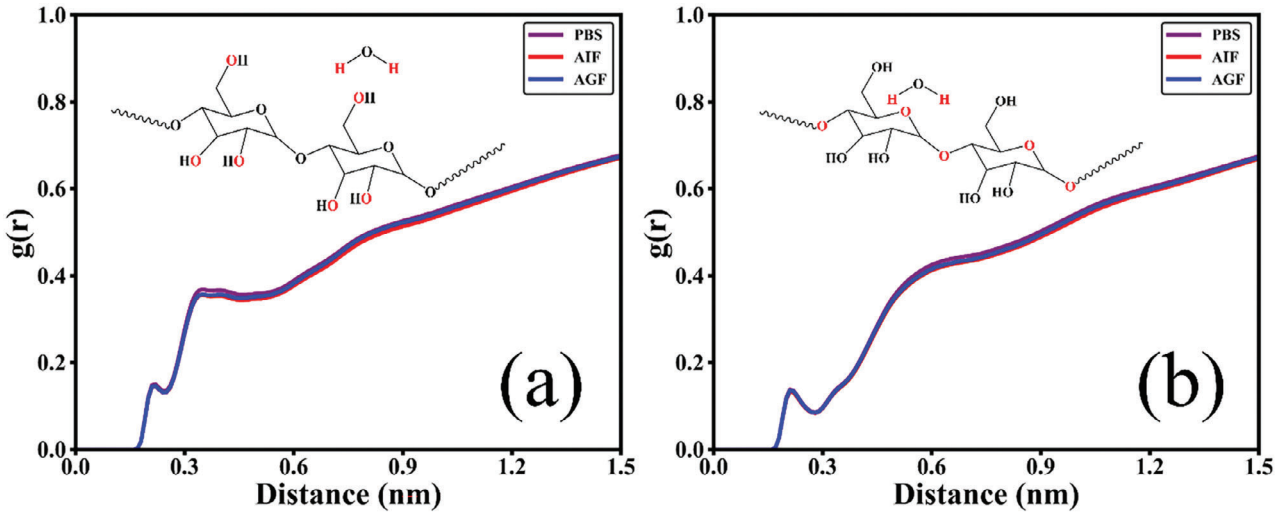


Figure 5. RDF between OSA-modified starch moiety to the hydrogen of water a) oxygen of hydroxyl moiety of OSA-modified starch to the H of water b) ether moiety present in the OSA-modified starch to the H of water.

origins. The slope of the MSD versus time plot provides information about the diffusion coefficient of the particles, allowing researchers to study dynamic properties such as molecular mobility and phase transitions.[49]

3. Result and Discussion

3.1. Identifying Drug-Binding Site

The docking results of lutein with gelatin and lutein with OSA-modified starch are provided in **Tables 2** and **3**, **Figure 2**, and Figures S1 and S2 (Supporting Information). The docking scores indicate a higher binding affinity of lutein for OSA-modified starch (−4.2 kcal mol^{-1}) compared to gelatin (−3.2 kcal mol^{-1}). This difference in binding affinity suggests that lutein has a stronger interaction with the OSA-modified starch component of the hydrogel. The higher docking score for OSA-modified starch implies that lutein molecules are more likely to associate with and be retained by the OSA-modified starch portions of the hydrogel. This high docking score observed due to the OSA modification introduces hydrophobic octenyl groups to the starch molecule, creating a more favorable environment for the hydrophobic regions of lutein. This hydrophobic interaction likely contributes significantly to the stronger binding. Additionally, the branched structure of OSA-modified starch may provide more accessible binding sites for lutein molecules. The presence of succinic acid groups in OSA-modified starch could also facilitate hydrogen bonding with lutein's hydroxyl groups. In contrast, gelatin's primarily hydrophilic nature and linear structure may offer fewer optimal binding sites for lutein, resulting in a lower binding affinity. While gelatin still shows some level of interaction with lutein, it appears to be less effective in binding and retaining the lutein molecules compared to OSA-modified starch.

3.2. Hydrogel-Drug Stability

Figure 3 shows the H-bonding between the polymer framework to the solvents. Figure 3a shows the H-bonding between the gelatin polymer framework to the water moiety. The average number of H-bonding between the gelatin polymer framework and different systems is increasing with respect to time. It suggests the dissociation of gelatin polymer frameworks. Initially, the gelatin framework shows the highest H-bonding in the AGF system. However, as the simulation progresses, the gelatin framework forms more H-bonds with PBS solution. This suggests the gelatin frameworks do not maintain their structural integrity compared to the AIF and AGF solutions. However, gelatin forms a smaller number of H-bonds in AIF solution due to the environmental effects. Figure 3b shows the H-bonding between the OST-modified starch to the solutions. The average number of H-bonding is shown to be constant throughout dynamics. This suggests that the OSA-modified starch maintains the structural integrity of the solution which is suggested by the experimental literature.[19] The order of the number of H-bonding is PBS>AGF>AIF. The solvent environment at different pH levels can influence the interactions between the polymer framework and the gastro-intestinal buffers. In the PBS system, the ions interact with the polymer frameworks. As a result, the polymer structure rearranges itself, which makes them more hydrophilic in nature. Combining Figure 2a,b, the gelatin helps to form a hydrogel network which is suggested by the experimental literature.[19] However, the OST-modified starch provides the stability of hydrogel networks which is suggested by the experimental literature.[19] The interactions between the polymer moiety and solutions are given in further sections.

Figure S3a (Supporting Information) of SM shows the H-bonding between lutein to the OST-modified starch. The order of the number of H-bonding is AGF>AIF>PBS. In the AGF solution, the OST starch holds its structural integrity, and it holds drug molecules for a long time. As the pH of the system increases, the hydrophilicity of starch increases, which is due to the interaction between the ions present in the solution and the polymer. As the hydrophilicity of OST starch increases, the drug-holding capacity also decreases. Figure S3b (Supporting Information) shows the H-bonding between the lutein to the gastro-intestinal buffers. It is observed that the PBS medium shows

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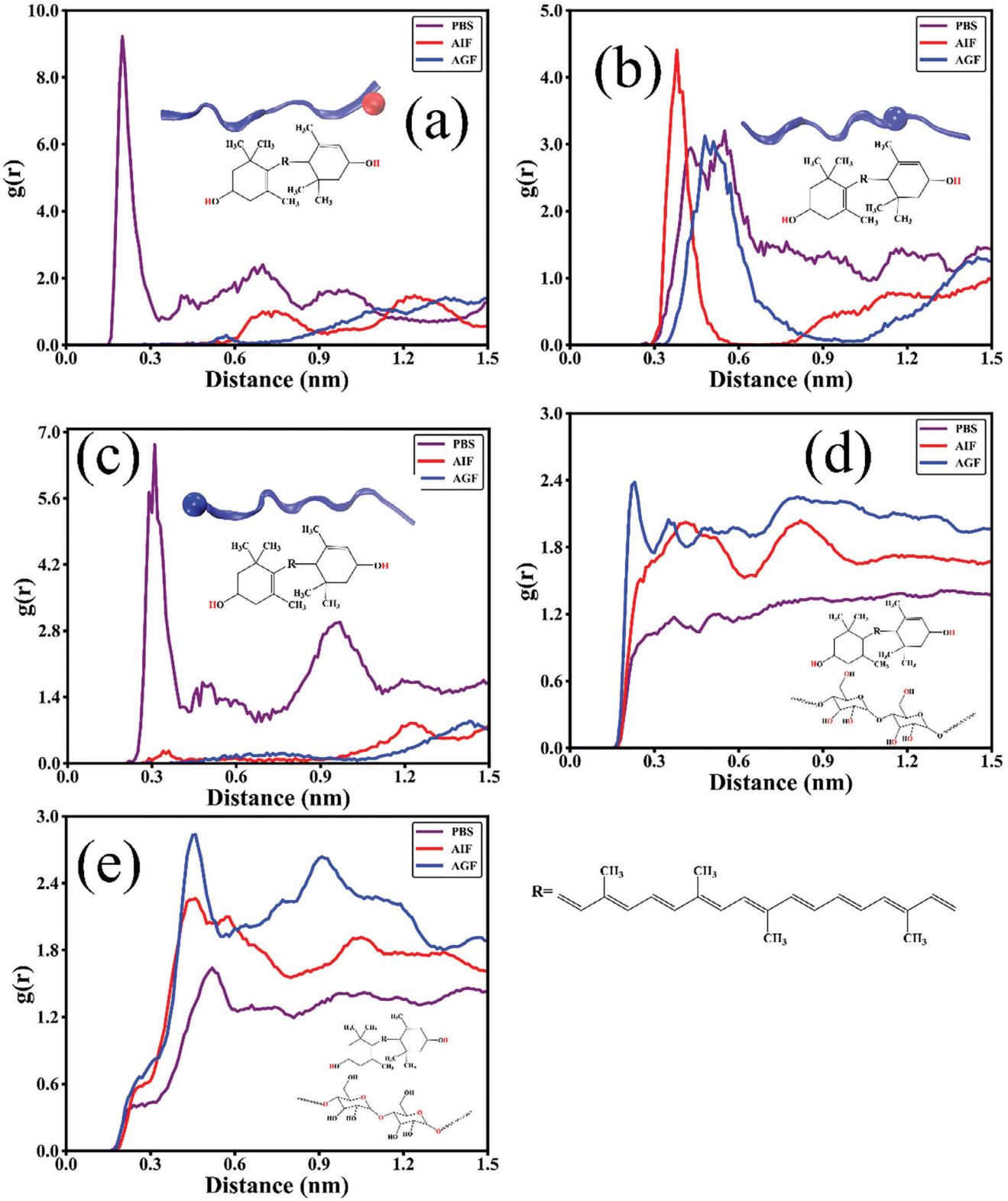


Figure 6. RDF between the H of hydroxyl moiety of lutein to the other moiety of polymer a) O of alanine to the H of hydroxyl moiety of lutein b) N of arginine to the H of hydroxyl moiety of lutein c) N of lysine to the H of hydroxyl moiety of lutein d) O of hydroxyl of OSA-modified starch to the H of hydroxyl moiety of lutein e) ether of OST modified starch to the H of hydroxyl moiety of lutein.

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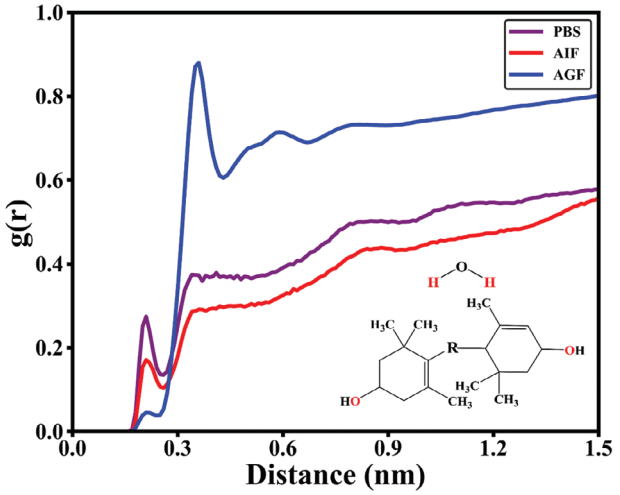


Figure 7. RDF of O of hydroxyl moiety presents in lutein to the H of water.

a higher amount of H-bonding than other mediums. In a PBS medium, the polymer forms more H-bonds with a solution. As a result, the active site of H-bonding is reduced, which can influence the release of the drug moiety from polymer composites to solution mediums. However, the gelatin framework does not form an H-bonding with lutein drug moiety.

3.3. Hydrogel-Drug-Buffer Interactions

Figure 4a–l shows the RDF between the different moieties of gelatin to the hydrogen moiety of water. The first solvation shell is observed ≈0.3 nm except the Figure 4a–j. The corresponding CN is provided in Table S1 (Supporting Information). The first solvation shell is observed in hydrogen bonding regions, which suggests gelatin is a very good material for forming a hydrogel framework which is suggested in the previous literature.[19,50,51] Figure 4a–j shows the first solvation shell ≈0.44 nm, which suggests the strong vdW interaction between the moieties which is suggested in the previous literature.[19,25,50] However, in Figure 4d, the RDF curve between the chlorine moiety and the hydrogen moiety of water shows stronger interactions in AGF media compared to the PBS and AIF media. The AGF media contains H_3O^+ ions which enhance the RDF peak with water. **Figure** 5a,b shows the RDF between the oxygen of hydroxyl moiety of OSA-modified starch to the hydrogen moiety of water and the ether moiety present in the OSA-modified starch to the hydrogen moiety of water, respectively. The first solvation shell is observed ≈0.30 with a CN of 0.03 and 0.08, respectively. The first solvation shell is observed in the hydrogen bonding regions which is suggested in the previous literature.[19,49,51] The RDF between the hydrogen moiety of water to the different moiety of gelatin and different moiety OSA-modified starch shows that both components are capable to form a hydrogel frame work which is also discussed in our previous section. However, the RDF peak shows stronger interactions with gelatins compared to the OSA-modified starch which also validates the hydrogen bonding results.

The RDF result of the previous section suggests the formation of a hydrogel framework, and the hydrogel carries the drug molecules into the targeted site. To understand the drug carrying mechanism of lutein, the interactions between the lutein and different moiety of the hydrogel framework are necessary, which is shown in Figure 5. **Figure** 6a–c shows the RDF between the hydrogen of the hydroxyl moiety of lutein to the different moiety of gelatin framework in the AGF, AIF, and PBS media. In Figure 6a, the first solvation shell is observed at 0.32 nm for PBS media, 0.64 nm for AGF media, and 0.9 for the AIF. The first solvation shell is observed at PBS media in the hydrogen bonding regions which is suggested in the previous literature.[19,49,51] However, in AIF and AGF media first solvation is observed due to the weak electrostatic interactions between the molecules. In Figure 6b, the first solvation shell is observed at 0.6 nm for AIF, 0.78 nm PBS, and 0.93 nm for AGF media respectively. The first solvation shell is observed due to the weak electrostatic interactions between the molecules which is suggested in the previous literature.[19,25,50] In Figure 6c, the first solvation shell is observed at 0.40 nm for PBS and AIF media and 0.62 for AGF media. The first solvation shell observed weak electrostatic interactions between the moieties. Summing up Figure 6a–c, the alanine and lysine amino acids show stronger interaction in PBS media, and arginine shows stronger interaction in the AGF media. Lysine is positively charged at physiological pH, allowing it to form ionic bonds with lutein, which enhances binding. Alanine, being hydrophobic, contributes through non-polar interactions. Due to the hydrophobicity of Alanine, it is forming strong interactions with lutein compared to the other amino acids. Figure 6d–e shows the RDF between the hydrogen of hydroxyl moiety of lutein to the oxygen of hydroxyl moiety and ether moiety present in OSA-modified starch, respectively. In Figure 6d, the first solvation shell is observed at 0.3 nm for the AGF system, 0.46 nm for PBS, and 0.6 nm for the AIF system with a CN of 0.70, 0.01, and 1.87, respectively. In AGF media, the first solvation shell is in the hydrogen bonding regions. However, in the AIF and PBS media, the solvation shell is observed due to the vdW interactions among the molecules. Figure 6e shows the RDF between the ether moiety of OST-modified starch to the hydrogen of hydroxyl moiety of lutein. The first solvation shell is observed at 0.57 nm with a CN 2.87, 4.46, and 4.71, respectively. The solvation shell is observed due to the vdW interactions among the molecules. Summing up Figure 6a–e, the lutein has a strong interaction with the OSA-modified starch compared to the gelatin. However, gelatin only interacts in a PBS medium, and as pH goes to acidic media, the gelatin does not interact with the lutein molecules. **Figure** 7 shows the RDF between the oxygen of the hydroxyl group of lutein to the hydrogen moiety of water. The first solvation shell is observed at 0.30 nm with a CN of 0.10, 0.11, and 0.08, respectively. The first solvation shell is observed in the hydrogen bonding regions. This suggests that lutein is also interacting with water.

3.4. Dynamic Drug Displacement

Figure 9 shows the MSD of oxygen of the hydroxyl group of lutein in PBS, AIF, and AGF media. In Figure 9a, the MSD shows linear behavior till 35 ns in PBS media. During this time period, the logarithmic slope of MSD is less than one which signifies the subdiffusive behavior which is suggested in the previous literature.[52] This behavior is observed due to the drug molecules sticking into the polymer pocket. The polymer matrix resists the free diffusion

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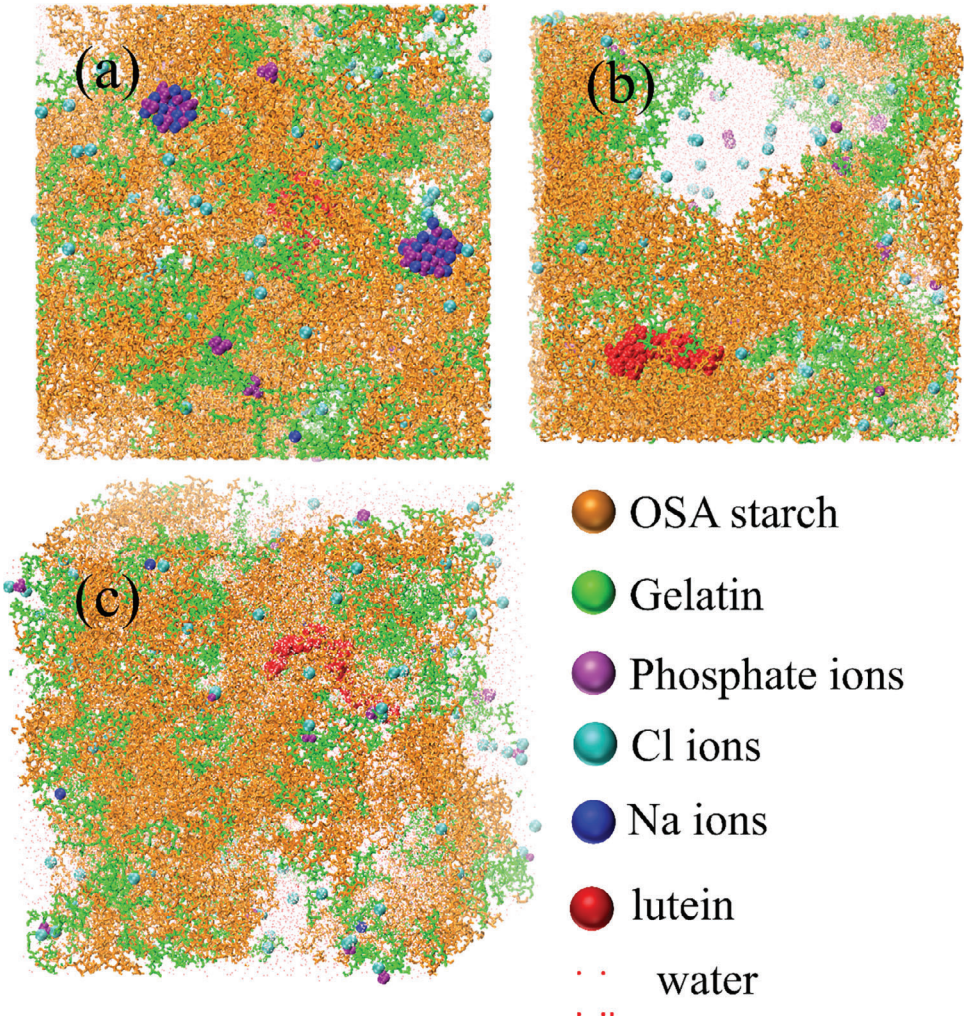


Figure 8. Simulation snapshot at 50 ns a) PBS b) AIF c) AGF.

of drug moiety. After that, it shows hindered diffusive behavior till the end of the dynamics. During this time period the drug molecules try to break the binding pocket in a polymer matrix. In Figure 9b the MSD of lutein is shown in AIF media. Initially, the MSD curve shows linear behavior 22 ns. During this time periods, the logarithmic slope value is less than one. It signifies the subdiffusive behavior which is suggested in the previous literature.[52] This behavior is caused by the drug molecules sticking into polymer pockets. After that, the MSD curve shows an increasing concave nature till the end of the dynamics. During this time period, the hydrophilic part of the drug moiety interacts with solvents and the hydrophobic part stays in the polymer matrix. Figure 9c shows the MSD of lutein in the AIF media. Initially, the MSD shows linear behavior till 30 ns. During this time periods, the logarithmic slope value is less than one which signifies the subdiffusive behavior. This is suggested in the previous literature.[52] This subdiffusive behavior is observed due to the polymer moiety sticking into the binding pocket of the polymer which is shown in **Figure 8**. After that, the MSD shows a negative slope till the end of the dynamics. During this time period the motion of the drug molecules hindered by the polymer matrix. Comparing **Figure 9a–c**, the highest MSD of lutein is observed in a PBS medium. It signifies the drug molecules diffuse from the polymer matrix to the solution matrix faster in the PBS media.

4. Conclusion

The all-atom molecular dynamics simulations are used to understand the dynamics of lutein within the macromolecules, which contain gelatin and OSA-modified starch, submerged into the different gastro-intestinal buffers. The H-bonding results suggest that the gelatin and OSA starch both are able to form a hydrogel network. As time progresses, the gelatin enhances the H-bond with water which signifies the loss of structural integrity faster. However, the OSA starch maintains its structural integrity in all gastro-intestinal buffers. The RDF results between gelatin moiety to water moiety are higher compared to the OSA starch moiety to the water moiety. The H-bonding results also validate this finding. To understand the binding of lutein in different polymer frameworks, the auto dock is used. The autodock score suggests the OSA-starch has higher binding affinity compared to the

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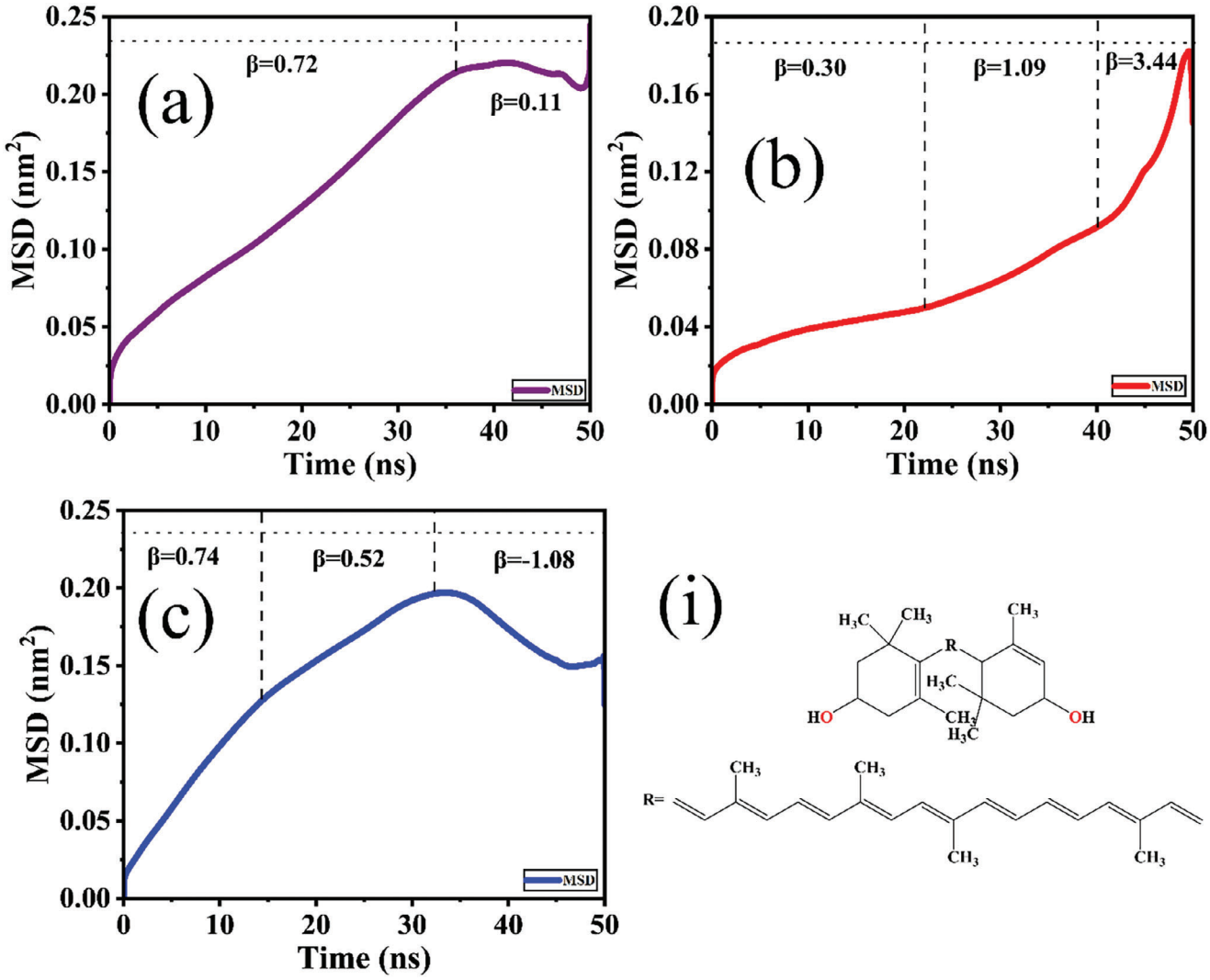


Figure 9. MSD of oxygen of hydroxyl present in the lutein a) PBS b) AIF c) AGF (i) schematic representations of lutein.

gelatin. Further validate this result, the H-bonding and RDF are plotted. The H-bonding result suggest that the lutein is only capable of forming H-bond with OSA-starch but not the gelatin. The RDF result between lutein and other moiety suggests that the major interactions are vdW and electrostatic interactions. However, the hydrophilic part of the lutein interacts with water, which is validated by the H-bonding and RDF results. Further, to understand the permeation of lutein, the MSD is plotted. The MSD result shows the subdiffusive behavior of lutein in three buffers. However, the MSD of lutein shows higher in PBS media, which suggests a higher self-diffusive coefficient value compared to the other media. Overall, these results are elaborating the understanding of interactions of lutein with solvents as well as components of hydrogels. These findings are expected to contribute valuable insights for the rational design of improved lutein delivery systems, potentially enhancing bioavailability and stability in nutraceutical and pharmaceutical applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

gastro-intestinal buffers, gelatin, lutein, molecular dynamics simulations, octenyl succinic anhydride-modified starch

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